Restriction of Scalars and Proper Base Change

A Detailed Derived-Categorical Proof

Abstract

Let $A \to B$ be a homomorphism of commutative rings, let $F \in \mathrm{D}^+(X_{\text{ét}}, B)$, and consider the canonical proper-base-change morphism with B-coefficients. Suppose that, after forgetting the B-module structure, the corresponding canonical morphism with A-coefficients is known to be an isomorphism. This note proves in detail that the original B-coefficient morphism is an isomorphism as well. The proof does not repeat proper base change. Its ingredients are the biadjunction

$$B_Z \otimes_{A_Z} - \dashv \mathrm{Res}_{B/A,Z} \dashv \underline{\mathrm{Hom}}_{A_Z}(B_Z, -),$$

which makes restriction of scalars exact, its separate faithfulness and conservativity, the composition theorem for right derived functors applied to the underived identity $\mathrm{Res}\, u_*^B = u_*^A \mathrm{Res}$, the one acyclicity check needed by that theorem, and the resulting canonical comparison

$$\mathrm{Res}_{B/A}\, \mathrm{R}u_*^B \xrightarrow{\sim} \mathrm{R}u_*^A\, \mathrm{Res}_{B/A},$$

The comparison is compatible with the units and counits of the A- and B-coefficient inverse/direct-image adjunctions, and with the naturality of the mate construction defining the base-change morphism. Particular care is taken not to assert that restriction of scalars preserves injective or K-injective objects.

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1 The precise question

Let $A \to B$ be a homomorphism of commutative rings. Consider a Cartesian square of schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y. \end{array} \tag{1.1}$$

For a scheme Z, write A_Z and B_Z for the constant sheaves of rings on $Z_{\text{ét}}$, and write

$$\mathrm{D}^+(Z_{\text{ét}}, A) = \mathrm{D}^+(\mathrm{Mod}(A_Z)), \qquad \mathrm{D}^+(Z_{\text{ét}}, B) = \mathrm{D}^+(\mathrm{Mod}(B_Z)).$$

We use a superscript A or B on a direct image when it is useful to record the coefficient category in which the derived functor is taken. For constant coefficients, inverse image is exact, and we write u^* instead of $\mathrm{L}u^*$.

For $F \in \mathrm{D}^+(X_{\text{ét}}, B)$, the canonical B-coefficient exchange map is

$$\beta_B(F) : g^*\mathrm{R}f_*^B F \longrightarrow \mathrm{R}f'^B_* g'^* F. \tag{1.2}$$

The question is whether one can prove that (1.2) is an isomorphism from the fact that the A-coefficient map

$$\beta_A(\mathrm{Res}_X F) : g^*\mathrm{R}f_*^A(\mathrm{Res}_X F) \longrightarrow \mathrm{R}f'^A_* g'^*(\mathrm{Res}_X F) \tag{1.3}$$

is an isomorphism. Here

$$\mathrm{Res}_Z = \mathrm{Res}_{B/A,Z} : \mathrm{Mod}(B_Z) \longrightarrow \mathrm{Mod}(A_Z)$$

denotes restriction of scalars.

The answer is yes, provided that (1.3) is the *canonical* base-change mate and not merely some unrelated A-linear isomorphism between the same two objects. The exact statement is the following.

Theorem 1.1 (Coefficient-forgetting transfer)**.** *In the situation above, assume that the canonical map $\beta_A(\mathrm{Res}_X F)$ is an isomorphism. Then the canonical map $\beta_B(F)$ is an isomorphism.*

No flatness assumption on $A \to B$ is needed for this implication. All geometric hypotheses, such as properness of f, are used only to know that (1.3) *is an isomorphism; they are not used again in the formal coefficient-transfer argument.*

The proof occupies Sections 2–10.

2 Restriction of scalars on sheaves

We begin at the abelian level. The cleanest proof of exactness is not a separate kernel-and-cokernel calculation: it is the fact that restriction of scalars has an adjoint on each side.

2.1 The three coefficient functors

The map $A \to B$ induces a map of constant ring sheaves $A_Z \to B_Z$. Write

$$\mathsf{U}_Z = \mathrm{Res}_{B/A,Z} : \mathrm{Mod}(B_Z) \longrightarrow \mathrm{Mod}(A_Z).$$

For a B_Z-module sheaf M, the object $\mathsf{U}_Z M$ is the same sheaf of abelian groups, with A_Z acting through $A_Z \to B_Z$. The functor changes neither sections nor the underlying map of a morphism.

There are also functors

$$\mathcal{E}_Z(N) = B_Z \otimes_{A_Z} N, \qquad \mathcal{C}_Z(N) = \underline{\mathrm{Hom}}_{A_Z}(B_Z, N) \quad (N \in \mathrm{Mod}(A_Z)). \tag{2.1}$$

Here $\underline{\mathrm{Hom}}$ is the internal sheaf Hom. If $V \to Z$ is an etale object, a section of $\mathcal{C}_Z(N)$ over V is an A_V-linear map $B_V \to N_V$, where $A_V = A_Z|_V$ and similarly for B_V, N_V. Its B_V-action is

$$(b \cdot \phi)(b') = \phi(b'b). \tag{2.2}$$

These formulas commute with restriction to smaller etale objects, so they define a sheaf of B_Z-modules.

Lemma 2.1 (Biadjunction and exactness)**.** *There are natural adjunctions*

$$\boxed{\mathcal{E}_Z = B_Z \otimes_{A_Z} - \ \dashv\ \mathsf{U}_Z \ \dashv\ \mathcal{C}_Z = \underline{\mathrm{Hom}}_{A_Z}(B_Z, -).} \tag{2.3}$$

Consequently, U_Z *is exact for every ring homomorphism* $A \to B$*.*

Proof. We first prove the left adjunction. For $N \in \mathrm{Mod}(A_Z)$ and $M \in \mathrm{Mod}(B_Z)$, define

$$\begin{aligned} \mathrm{Hom}_{B_Z}(B_Z \otimes_{A_Z} N, M) &\longrightarrow \mathrm{Hom}_{A_Z}(N, \mathsf{U}_Z M), \\ \Phi &\longmapsto (n \longmapsto \Phi(1 \otimes n)). \end{aligned} \tag{2.4}$$

Conversely, an A_Z-linear map $h : N \to \mathsf{U}_Z M$ defines

$$\widetilde{h} : B_Z \otimes_{A_Z} N \longrightarrow M, \qquad b \otimes n \longmapsto b\, h(n). \tag{2.5}$$

The balancing relation follows from A_Z-linearity:

$$\widetilde{h}(ba \otimes n) = ba\, h(n) = b\, h(an) = \widetilde{h}(b \otimes an).$$

The map $\widetilde{h}$ is B_Z-linear, and (2.4)–(2.5) are mutually inverse. Their formulas commute with every morphism in M and N, so the bijection is natural. This proves $\mathcal{E}_Z \dashv \mathsf{U}_Z$.

For the right adjunction, let $h : \mathsf{U}_Z M \to N$ be A_Z-linear. Define

$$\widehat{h} : M \longrightarrow \mathcal{C}_Z(N), \qquad \widehat{h}(m)(b) = h(bm). \tag{2.6}$$

It is B_Z-linear because, using (2.2),

$$\widehat{h}(b_0 m)(b) = h(bb_0 m) = (b_0 \cdot \widehat{h}(m))(b).$$

Conversely, if $k : M \to \mathcal{C}_Z(N)$ is B_Z-linear, evaluation at the unit gives an A_Z-linear map

$$\check{k} : \mathsf{U}_Z M \longrightarrow N, \qquad \check{k}(m) = k(m)(1). \tag{2.7}$$

Indeed, for a local section a of A_Z, its image in B_Z satisfies

$$\check{k}(am) = k(am)(1) = (a \cdot k(m))(1) = k(m)(a) = a\,k(m)(1).$$

Finally,

$$\check{\widehat{h}}(m) = \widehat{h}(m)(1) = h(m), \qquad \widehat{\check{k}}(m)(b) = \check{k}(bm) = k(bm)(1) = k(m)(b).$$

Thus the two constructions are inverse and natural, proving $\mathsf{U}_Z \dashv \mathcal{C}_Z$.

Now exactness is formal. Since U_Z is a right adjoint to $\mathcal{E}_Z$, it preserves all limits, hence in particular kernels; therefore it is left exact. Since U_Z is also a left adjoint to $\mathcal{C}_Z$, it preserves all colimits, hence in particular cokernels; therefore it is right exact. The functor is additive, so being both left and right exact means that it is exact. □

Lemma 2.2 (Faithfulness and conservativity)**.** *The functor $\mathsf{U}_Z : \mathrm{Mod}(B_Z) \to \mathrm{Mod}(A_Z)$ is faithful and conservative. In particular, if $\mathsf{U}_Z(\varphi)$ is an isomorphism, then φ is an isomorphism.*

Proof. Faithfulness is immediate from the definition: U_Z does not change the underlying morphism of sheaves. Thus $\mathsf{U}_Z(\varphi) = \mathsf{U}_Z(\psi)$ implies $\varphi = \psi$.

We record the standard implication "exact and faithful implies conservative." First, a faithful functor reflects zero objects. If $\mathsf{U}_Z Q = 0$, then

$$\mathsf{U}_Z(\mathrm{id}_Q) = \mathrm{id}_{\mathsf{U}_Z Q} = 0;$$

faithfulness gives $\mathrm{id}_Q = 0$, hence $Q = 0$. Now suppose that $\mathsf{U}_Z(\varphi)$ is an isomorphism. Exactness from Lemma 2.1 gives

$$\mathsf{U}_Z(\ker \varphi) = \ker(\mathsf{U}_Z \varphi) = 0, \qquad \mathsf{U}_Z(\mathrm{coker}\, \varphi) = \mathrm{coker}(\mathsf{U}_Z \varphi) = 0.$$

Reflection of zero objects yields $\ker \varphi = 0$ and $\mathrm{coker}\, \varphi = 0$. Hence φ is an isomorphism in the abelian category $\mathrm{Mod}(B_Z)$.

Equivalently, if ψ is the inverse of the underlying A_Z-linear map, then for local $b \in B_Z$ and $n \in N$, injectivity of φ and B_Z-linearity give

$$\varphi(b\psi(n)) = bn = \varphi(\psi(bn)),$$

so $\psi(bn) = b\psi(n)$. Thus the inverse is already B_Z-linear. □

Remark 2.3 (What the adjunctions do and do not prove)**.** *The two adjunctions prove exactness, but an arbitrary functor having adjoints on both sides need not be conservative. Conservativity here uses the additional concrete fact that restriction of scalars is faithful, indeed that it leaves the underlying sheaf unchanged.*

No flatness of $A \to B$ is involved. Flatness would make the left adjoint $B_Z \otimes_{A_Z} -$ exact; it is not needed to make the middle functor U_Z exact. The middle functor is exact because it is simultaneously a left and a right adjoint.

3 Restriction of scalars on derived categories

The derived statement is now a formal consequence of the abelian biadjunction. Because U_Z is exact, no resolution is needed to derive it: apply it degree by degree.

Proposition 3.1 (Formal passage to the derived category)**.** *Termwise restriction of scalars defines an exact triangulated functor*

$$\mathsf{U}_Z : \mathrm{D}^+(Z_{\text{ét}}, B) \longrightarrow \mathrm{D}^+(Z_{\text{ét}}, A).$$

It is the total derived functor of U_Z *on either side,*

$$\mathrm{L}\mathsf{U}_Z \simeq \mathsf{U}_Z \simeq \mathrm{R}\mathsf{U}_Z, \tag{3.1}$$

and for every K *and every integer* n *there is a natural isomorphism*

$$H^n(\mathsf{U}_Z K) \xrightarrow{\sim} \mathsf{U}_Z H^n(K). \tag{3.2}$$

In particular, U_Z *is* t*-exact.*

Proof. Let $K^\bullet$ be a complex of B_Z-modules. Exactness of U_Z gives

$$\mathsf{U}_Z(\ker d^n) = \ker(\mathsf{U}_Z d^n), \qquad \mathsf{U}_Z(\operatorname{im} d^{n-1}) = \operatorname{im}(\mathsf{U}_Z d^{n-1}).$$

Since an exact functor also preserves the cokernel defining the quotient of a kernel by an image,

$$\begin{aligned} H^n(\mathsf{U}_Z K) &= \ker(\mathsf{U}_Z d^n)/\operatorname{im}(\mathsf{U}_Z d^{n-1}) \\ &\cong \mathsf{U}_Z\big(\ker d^n/\operatorname{im} d^{n-1}\big) = \mathsf{U}_Z H^n(K). \end{aligned}$$

This proves (3.2). In particular, U_Z sends acyclic complexes to acyclic complexes and quasi-isomorphisms to quasi-isomorphisms. Hence its termwise functor on the homotopy category descends to the localization defining the derived category. It commutes termwise with shifts and mapping cones, so the descended functor is triangulated. Formula (3.2) says exactly that it preserves both halves of the standard t-structure, hence is t-exact. Finally, an exact functor is already its own left and right derived functor, which gives (3.1). □

Proposition 3.2 (The right adjunction derives on D^+)**.** *Let* $\mathcal{C}_Z = \underline{\mathrm{Hom}}_{A_Z}(B_Z, -)$*. There is a derived adjunction*

$$\mathsf{U}_Z : \mathrm{D}^+(Z_{\text{ét}}, B) \rightleftarrows \mathrm{D}^+(Z_{\text{ét}}, A) : \mathrm{R}\mathcal{C}_Z, \qquad \mathsf{U}_Z \dashv \mathrm{R}\mathcal{C}_Z. \tag{3.3}$$

Proof. The categories of module sheaves have enough injectives. Moreover, $\mathcal{C}_Z$ sends injective A_Z-modules to injective B_Z-modules. To see this, let I be injective in $\mathrm{Mod}(A_Z)$ and let $M' \hookrightarrow M$ be a monomorphism in $\mathrm{Mod}(B_Z)$. Exactness of U_Z makes $\mathsf{U}_Z M' \hookrightarrow \mathsf{U}_Z M$ a monomorphism. The lifting property of I, followed by the abelian adjunction $\mathsf{U}_Z \dashv \mathcal{C}_Z$, shows that

$$\mathrm{Hom}_{B_Z}(M, \mathcal{C}_Z I) \longrightarrow \mathrm{Hom}_{B_Z}(M', \mathcal{C}_Z I)$$

is surjective. Hence $\mathcal{C}_Z I$ is injective.

Now resolve $L \in \mathrm{D}^+(Z_{\text{ét}}, A)$ by a bounded-below complex $L \to I^\bullet$ of injectives. Then $\mathcal{C}_Z(I^\bullet)$ is a bounded-below complex of injective B_Z-modules, so it represents $\mathrm{R}\mathcal{C}_Z(L)$. Applying the termwise abelian adjunction to complexes and then passing to homotopy classes gives, naturally in $K \in \mathrm{D}^+(Z_{\text{ét}}, B)$ and L,

$$\mathrm{Hom}_{\mathrm{D}^+(A_Z)}(\mathsf{U}_Z K, L) \cong \mathrm{Hom}_{\mathrm{D}^+(B_Z)}(K, \mathrm{R}\mathcal{C}_Z(L)).$$

This is (3.3). □

Remark 3.3 (The left derived adjunction and finite Tor amplitude)**.** *The left abelian adjunction $\mathcal{E}_Z \dashv \mathsf{U}_Z$ also derives, in a derived category admitting K-flat resolutions, to*

$$B_Z \otimes^{\mathrm{L}}_{A_Z} - \ \dashv\ \mathsf{U}_Z. \tag{3.4}$$

There is, however, a boundedness issue if one insists that both sides of (3.4) *lie in* D^+*.*

Assume that B_Z has Tor amplitude contained in $[-d, 0]$ over A_Z; that is,

$$\mathrm{Tor}_p^{A_Z}(B_Z, N) = 0 \qquad \text{for every } N \in \mathrm{Mod}(A_Z) \text{ and every } p > d. \tag{3.5}$$

For example, this holds if B_Z admits a flat A_Z-module resolution of length d. If $K \in \mathrm{D}^{\geq m}(Z_{\text{ét}}, A)$, the hyper-Tor spectral sequence

$$E_2^{-p,q} = \mathrm{Tor}_p^{A_Z}(B_Z, H^q(K)) \Longrightarrow H^{q-p}(B_Z \otimes^{\mathrm{L}}_{A_Z} K) \tag{3.6}$$

has $q \geq m$ and, by (3.5)*, $0 \leq p \leq d$. Therefore $q - p \geq m - d$, and*

$$B_Z \otimes^{\mathrm{L}}_{A_Z} K \in \mathrm{D}^{\geq m-d}(Z_{\text{ét}}, B).$$

Thus finite Tor amplitude guarantees that derived extension of scalars preserves bounded-below complexes and gives an adjunction entirely inside D^+*:*

$$B_Z \otimes^{\mathrm{L}}_{A_Z} - : \mathrm{D}^+(Z_{\text{ét}}, A) \rightleftarrows \mathrm{D}^+(Z_{\text{ét}}, B) : \mathsf{U}_Z. \tag{3.7}$$

When $d = 0$, extension of scalars is exact and this left adjoint is t-exact. Without a finite uniform Tor bound, there is in general no uniform lower bound for $B_Z \otimes^{\mathrm{L}}_{A_Z} N$ as N ranges over modules placed in degree zero; consequently $B_Z \otimes^{\mathrm{L}}_{A_Z} -$ need not carry D^+ *to* D^+*. This affects the availability of the* left adjoint *inside* D^+*, but it does not affect the exactness, t-exactness, or conservativity of the middle functor U_Z used in this note.*

Proposition 3.4 (Derived conservativity)**.** *The functor $\mathsf{U}_Z : \mathrm{D}^+(Z_{\text{ét}}, B) \to \mathrm{D}^+(Z_{\text{ét}}, A)$ is conservative. More precisely:*

1. *if $\mathsf{U}_Z K \simeq 0$, then $K \simeq 0$;*
2. *if $\alpha : K \to L$ and $\mathsf{U}_Z(\alpha)$ is an isomorphism, then α is an isomorphism.*

Proof. If $\mathsf{U}_Z K \simeq 0$, then t-exactness gives, for every n,

$$0 = H^n(\mathsf{U}_Z K) \cong \mathsf{U}_Z H^n(K).$$

Lemma 2.2 reflects zero objects, so $H^n(K) = 0$ for every n. An object of the derived category is zero exactly when all of its cohomology sheaves vanish. Thus $K \simeq 0$.

For a morphism α, form its distinguished triangle

$$K \xrightarrow{\alpha} L \longrightarrow \mathrm{Cone}(\alpha) \longrightarrow K[1].$$

Since U_Z is triangulated,

$$\mathsf{U}_Z \,\mathrm{Cone}(\alpha) \simeq \mathrm{Cone}(\mathsf{U}_Z \alpha).$$

If $\mathsf{U}_Z \alpha$ is an isomorphism, its cone is zero. The first part then gives $\mathrm{Cone}(\alpha) \simeq 0$, which is equivalent to α being an isomorphism. □

Remark 3.5. *The adjunctions are the efficient reason that restriction of scalars is exact, and exactness is the efficient reason that it is t-exact on derived categories. Conservativity still uses faithfulness on the heart. None of these arguments requires bounded-below complexes. The bounded-below hypothesis enters later, when ordinary injective resolutions are used to compare derived direct images.*

4 Compatibility with inverse and underived direct image

Let $u : Z_1 \to Z_0$ be a morphism of schemes. It induces a geometric morphism of etale topoi. We use

$$u_B^* : \mathrm{Mod}(B_{Z_0}) \rightleftarrows \mathrm{Mod}(B_{Z_1}) : u_*^B$$

and the analogous adjunction with A-coefficients.

Lemma 4.1 (Inverse image)**.** *There is a canonical natural isomorphism*

$$\lambda_u : \mathrm{Res}_{Z_1} u_B^* \xrightarrow{\sim} u_A^* \mathrm{Res}_{Z_0} . \tag{4.1}$$

The inverse-image functors u_A^* and u_B^* are exact.

Proof. The inverse-image functor of a geometric morphism is left exact by definition and is a left adjoint, hence preserves all colimits. In an abelian category, preservation of finite limits gives preservation of kernels, while preservation of colimits gives preservation of cokernels. Therefore it is exact on sheaves of abelian groups and on sheaves of modules over constant coefficient rings.

The construction of inverse image uses only inverse image of the underlying sheaf and sheafification. The B-action is transported along this construction. Forgetting the action before or after the construction gives the same A-module sheaf. Equivalently, at every geometric point $\bar{z}$ of Z_1, both sides of (4.1) have stalk equal to the same underlying A-module

$$M_{u(\bar{z})}.$$

The etale topos has enough geometric points, so the canonical comparison is an isomorphism. □

Lemma 4.2 (Underived direct image)**.** *There is a canonical equality of functors*

$$\mathrm{Res}_{Z_0} u_*^B = u_*^A \mathrm{Res}_{Z_1} . \tag{4.2}$$

Proof. For an etale object $V \to Z_0$ and a B_{Z_1}-module sheaf M,

$$(u_*^B M)(V) = M(Z_1 \times_{Z_0} V).$$

The same formula defines the underlying A-module sections of $u_*^A \mathrm{Res}_{Z_1} M$. Restriction maps and the A-action also agree. Thus the two functors are equal after the evident identification. □

5 The acyclicity input for derived composition

The natural way to derive the underived equality (4.2) is to use the composition theorem for right derived functors. We first recall the precise form that will be used.

Proposition 5.1 (Composition theorem for bounded-below right derived functors)**.** *Let*

$$\mathcal{C} \xrightarrow{G} \mathcal{D} \xrightarrow{F} \mathcal{E}$$

be left exact functors between abelian categories with enough injectives. Suppose that G sends injective objects of $\mathcal{C}$ to F-acyclic objects of $\mathcal{D}$. Then the canonical comparison is an isomorphism

$$\mathrm{R}F \circ \mathrm{R}G \xrightarrow{\sim} \mathrm{R}(F \circ G) \tag{5.1}$$

on bounded-below derived categories.

In particular:

1. *if F is exact, then every object is F-acyclic and*

$$F \circ \mathrm{R}G \xrightarrow{\sim} \mathrm{R}(F \circ G); \tag{5.2}$$

2. *if G is exact and sends injectives to F-acyclic objects, then*

$$\mathrm{R}F \circ G \xrightarrow{\sim} \mathrm{R}(F \circ G). \tag{5.3}$$

These isomorphisms are natural and compatible with natural transformations of the underived composite functors.

Proof. Let $K \to I^\bullet$ be a bounded-below injective resolution in $\mathcal{C}$. The complex $G(I^\bullet)$ represents $\mathrm{R}G(K)$. By hypothesis, every term $G(I^n)$ is F-acyclic. The acyclic-resolution theorem therefore computes

$$\mathrm{R}F(\mathrm{R}G(K)) \simeq F(G(I^\bullet)).$$

The same injective resolution computes the total right derived functor of the composite:

$$\mathrm{R}(F \circ G)(K) \simeq F(G(I^\bullet)).$$

These identifications give (5.1). They are independent of the chosen resolution and natural in K by the comparison theorem for resolutions.

If F is exact, then $\mathrm{R}F = F$ and every object is F-acyclic, giving (5.2). If G is exact, then $\mathrm{R}G = G$, and (5.1) gives (5.3). A natural transformation of underived composites induces a map of the complexes $F(G(I^\bullet))$; consequently the displayed comparison commutes with derived natural transformations. □

For the composite $\mathrm{Res}_{Z_0} \circ u_*^B$, the outer functor Res_{Z_0} is exact, so (5.2) applies immediately. For $u_*^A \circ \mathrm{Res}_{Z_1}$, formula (5.3) requires one additional statement: restriction of scalars must send a B_{Z_1}-injective sheaf to a u_*^A-acyclic A_{Z_1}-module sheaf. It need not send it to an injective sheaf. The rest of this section verifies exactly this weaker condition.

Lemma 5.2 (Cohomology does not depend on the recorded scalar ring)**.** *Let T be an etale topos and let M be a sheaf of B-modules on T. For every $q \geq 0$, there is a natural identification of underlying A-modules*

$$\mathrm{Res}_{B/A} H_B^q(T, M) \xrightarrow{\sim} H_A^q(T, \mathrm{Res}_{B/A} M). \tag{5.4}$$

The same statement holds on every slice etale topos.

Proof. We spell out why the coefficient category does not change the answer. An etale topos has enough geometric points. Choose a conservative family of points and form the functorial Godement resolution of the underlying sheaf of abelian groups:

$$0 \longrightarrow M \longrightarrow G^0(M) \longrightarrow G^1(M) \longrightarrow \cdots. \tag{5.5}$$

The construction takes stalks, products of skyscraper sheaves, and successive cokernels. It is functorial. Hence every endomorphism of M, in particular multiplication by every local section of B, extends functorially to every term and every differential of (5.5). Thus the same resolution is simultaneously a complex of B-module sheaves and, after forgetting, a complex of A-module sheaves.

Each $G^n(M)$ is flasque in the topos-theoretic sense relevant to this construction and is acyclic for global sections. Consequently both $H_B^q(T, M)$ and $H_A^q(T, \mathrm{Res}\, M)$ are computed by the same underlying complex of global sections

$$\Gamma(T, G^0(M)) \longrightarrow \Gamma(T, G^1(M)) \longrightarrow \cdots.$$

The cohomology groups are therefore the same abelian groups, and the A-action on both is obtained from the B-action through $A \to B$. This gives (5.4). The argument applies verbatim to a slice topos, which again has enough points. □

Lemma 5.3 (Restriction to an etale slice preserves injectives)**.** *Let $j : W \to Z$ be an etale morphism and let I be an injective sheaf of B_Z-modules. Then $I|_{W_{\text{ét}}}$ is injective as a sheaf of B_W-modules.*

Proof. The restriction functor

$$j^* : \mathrm{Mod}(B_Z) \longrightarrow \mathrm{Mod}(B_W)$$

for the localization of the etale topos has an extension-by-zero left adjoint $j_!$. For an etale localization, $j_!$ is exact: locally it extends a section on an object over W by the initial section away from that object, and kernels and cokernels are computed sectionwise before sheafification. A right adjoint to an exact functor preserves injective objects. Indeed, if $M' \hookrightarrow M$ is a monomorphism on $W_{\text{ét}}$, exactness gives a monomorphism $j_!M' \hookrightarrow j_!M$, and adjunction turns the extension property against this monomorphism into the extension property for j^*I. Thus j^*I is injective. □

Proposition 5.4 (The required direct-image acyclicity)**.** *Let $u : Z_1 \to Z_0$ be a morphism of schemes and let I be injective in* $\mathrm{Mod}(B_{Z_1})$*. Then* $\mathrm{Res}_{Z_1} I$ *is u_*^A-acyclic:*

$$R^q u_*^A(\mathrm{Res}_{Z_1} I) = 0 \qquad (q > 0). \tag{5.6}$$

Thus Res_{Z_1} *satisfies the acyclicity hypothesis in Proposition 5.1 for the composite $u_*^A \circ \mathrm{Res}_{Z_1}$.*

Proof. Take an etale object $V \to Z_0$ and put $W = Z_1 \times_{Z_0} V$. The standard local description of higher direct images says that $R^q u_*^A(\mathrm{Res}\, I)$ is the sheaf associated to the presheaf

$$V \longmapsto H_A^q(W_{\text{ét}}, (\mathrm{Res}\, I)|_W).$$

By Lemma 5.3, $I|_W$ is injective as a sheaf of B_W-modules. An injective object is acyclic for every left exact functor, in particular for global sections, so

$$H_B^q(W_{\text{ét}}, I|_W) = 0 \qquad (q > 0).$$

Coefficient independence from Lemma 5.2 gives

$$H_A^q(W_{\text{ét}}, (\mathrm{Res}\, I)|_W) \cong \mathrm{Res}\, H_B^q(W_{\text{ét}}, I|_W) = 0.$$

This holds for every etale $V \to Z_0$, so the associated higher direct image sheaf vanishes. This proves (5.6). □

Remark 5.5. *This proposition is the only place where the local cohomological argument enters. Its role is to verify the hypothesis of the derived composition theorem. It is not the definition of the comparison χ_u, and there is no need to reconstruct χ_u term by term.*

6 The derived direct-image comparison

We now construct the essential comparison isomorphism.

Proposition 6.1 (Forgetting coefficients commutes with $\mathrm{R}u_*$)**.** *For every $K \in \mathrm{D}^+(Z_{1,\text{ét}}, B)$, there is a natural isomorphism*

$$\chi_{u,K} : \mathrm{Res}_{Z_0} \mathrm{R}u_*^B K \xrightarrow{\sim} \mathrm{R}u_*^A \mathrm{Res}_{Z_1} K. \tag{6.1}$$

It is the right mate, under the two adjunctions, of the inverse-image comparison (4.1)*.*

Proof. Put $\mathsf{U}_0 = \mathrm{Res}_{Z_0}$ and $\mathsf{U}_1 = \mathrm{Res}_{Z_1}$. At the abelian level, Lemma 4.2 gives a canonical equality of left exact functors from $\mathrm{Mod}(B_{Z_1})$ to $\mathrm{Mod}(A_{Z_0})$:

$$\mathsf{U}_0 \circ u_*^B = u_*^A \circ \mathsf{U}_1. \tag{6.2}$$

We derive this single equality and identify its two derived expressions.

Because U_0 is exact, formula (5.2) gives

$$\mathsf{U}_0 \circ \mathrm{R}u_*^B \xrightarrow{\sim} \mathrm{R}(\mathsf{U}_0 \circ u_*^B). \tag{6.3}$$

Because U_1 is exact and sends injective B_{Z_1}-module sheaves to u_*^A-acyclic objects by Proposition 5.4, formula (5.3) gives

$$\mathrm{R}u_*^A \circ \mathsf{U}_1 \xrightarrow{\sim} \mathrm{R}(u_*^A \circ \mathsf{U}_1). \tag{6.4}$$

Applying the total right-derived-functor construction to (6.2) and using (6.3)–(6.4) yields the canonical chain

$$\boxed{\mathsf{U}_0 \mathrm{R}u_*^B \xrightarrow{\sim} \mathrm{R}(\mathsf{U}_0 u_*^B) = \mathrm{R}(u_*^A \mathsf{U}_1) \xrightarrow{\sim} \mathrm{R}u_*^A \mathsf{U}_1.} \tag{6.5}$$

Its value at K is $\chi_{u,K}$. This construction immediately gives naturality: for every $a : K \to L$, functoriality of the total right derived functor gives

$$\begin{array}{ccc} \mathrm{Res}\,\mathrm{R}u_*^B K & \xrightarrow{\chi_{u,K}} & \mathrm{R}u_*^A \,\mathrm{Res}\, K \\ {\scriptstyle \mathrm{Res}\,\mathrm{R}u_*^B(a)}\downarrow & & \downarrow{\scriptstyle \mathrm{R}u_*^A(\mathrm{Res}\, a)} \\ \mathrm{Res}\,\mathrm{R}u_*^B L & \xrightarrow{\chi_{u,L}} & \mathrm{R}u_*^A \,\mathrm{Res}\, L, \end{array} \tag{6.6}$$

without any further choice of resolutions.

It remains to record why this canonical derived-composite comparison is the right mate claimed in the proposition. Let

$$L_B = u_B^*, \quad R_B = \mathrm{R}u_*^B, \qquad L_A = u_A^*, \quad R_A = \mathrm{R}u_*^A,$$

The strict inverse-image comparison is

$$\lambda_u : \mathsf{U}_1 L_B \xrightarrow{\sim} L_A \mathsf{U}_0.$$

At the abelian level, the right mate of this equality is exactly the sectionwise identity (6.2). The derived adjunctions are obtained by deriving the underived adjunctions, and the composition isomorphisms of Proposition 5.1 are compatible with natural transformations. Therefore deriving the underived right mate and then using the two composition isomorphisms gives the right mate of the derived left-adjoint comparison. Explicitly, that right mate is

$$\begin{aligned} \mathsf{U}_0 R_B \xrightarrow{\eta^A} R_A L_A \mathsf{U}_0 R_B \xrightarrow{R_A(\lambda_u^{-1})} R_A \mathsf{U}_1 L_B R_B \\ \xrightarrow{R_A \mathsf{U}_1(\epsilon^B)} R_A \mathsf{U}_1, \end{aligned} \tag{6.7}$$

where η^A is the unit of $L_A \dashv R_A$ and ϵ^B is the counit of $L_B \dashv R_B$. By the preceding derived-composition argument, this composite is precisely the map (6.5). Hence χ_u is simultaneously the canonical derived-composite isomorphism and the right mate of λ_u. □

Warning 6.2 (What was not proved or used)**.** *We did* not *claim that* $\mathrm{Res}\, I$ *is injective whenever* I *is injective, and we did not claim that restriction of scalars preserves* K*-injectives. We proved only the precise fact needed here:* $\mathrm{Res}\, I$ *is* u_*^A*-acyclic. This distinction is important when* $A \to B$ *is arbitrary.*

7 Units, counits, and the mate identities

Let η^A, ϵ^A and η^B, ϵ^B denote the units and counits of

$$u_A^* \dashv \mathrm{R}u_*^A, \qquad u_B^* \dashv \mathrm{R}u_*^B.$$

Because χ_u is the right mate of λ_u, it satisfies two identities. We record them with all objects indicated.

Proposition 7.1 (Unit identity)**.** *For every $M \in \mathrm{D}^+(Z_{0,\text{ét}}, B)$, after using $\lambda_u : \mathrm{Res}\, u_B^* M \simeq u_A^* \mathrm{Res}\, M$, one has*

$$\chi_{u,u_B^* M} \circ \mathrm{Res}(\eta_M^B) = \eta^A_{\mathrm{Res}\, M}. \tag{7.1}$$

Both sides are maps

$$\mathrm{Res}\, M \longrightarrow \mathrm{R}u_*^A u_A^* \mathrm{Res}\, M.$$

Proof. The unit $\eta_M^B : M \to R_B L_B M$ corresponds under $L_B \dashv R_B$ to $\mathrm{id}_{L_B M}$. Apply restriction of scalars and identify $\mathsf{U}_1 L_B M$ with $L_A \mathsf{U}_0 M$. By the definition of the right mate, postcomposing $\mathsf{U}_0 \eta_M^B$ with $\chi_{u,L_B M}$ gives the unique map whose adjoint under $L_A \dashv R_A$ is $\mathrm{id}_{L_A \mathsf{U}_0 M}$. That unique map is exactly the A-coefficient unit $\eta^A_{\mathsf{U}_0 M}$. This proves (7.1). □

Proposition 7.2 (Counit identity)**.** *For every $K \in \mathrm{D}^+(Z_{1,\text{ét}}, B)$, after the same inverse-image identification, one has*

$$\mathrm{Res}(\epsilon_K^B) = \epsilon^A_{\mathrm{Res}\, K} \circ u_A^*(\chi_{u,K}). \tag{7.2}$$

Both sides are maps

$$u_A^* \mathrm{Res}\, \mathrm{R}u_*^B K \longrightarrow \mathrm{Res}\, K.$$

Proof. Start with the right-mate formula (6.7). Apply L_A and then the A-counit. The triangle identity

$$\epsilon^A_{L_A N} \circ L_A(\eta_N^A) = \mathrm{id}_{L_A N}$$

cancels the inserted A-unit. The remaining composite is precisely the forgotten B-counit, transported through λ_u. This is (7.2).

Equivalently, ϵ_K^B corresponds to $\mathrm{id}_{R_B K}$ under the adjunction bijection. The defining property of mates says that transporting this identity through χ_u and λ_u gives $\epsilon^A_{\mathrm{Res}\, K}$. Both descriptions yield the same formula. □

Remark 7.3. *Equations* (7.1) *and* (7.2) *are stronger than the mere assertion that the two derived direct-image objects happen to be isomorphic. They say that the comparison is compatible with the adjunction data. This is exactly what is needed to compare canonical base-change maps, since those maps are built from a unit and a counit.*

8 The canonical base-change map as a mate

Return to the Cartesian square (1.1). There is a canonical isomorphism of exact inverse-image functors

$$\theta_R : f'^* g^* \xrightarrow{\sim} g'^* f^* \tag{8.1}$$

for either coefficient ring $R = A$ or $R = B$. Since all coefficient sheaves are constant, forgetting from B to A carries θ_B to θ_A literally.

Definition 8.1 (The exchange morphism)**.** *For $R \in \{A, B\}$ and $K \in \mathrm{D}^+(X_{\text{ét}}, R)$, define*

$$\beta_R(K) : g^*\mathrm{R}f_*^R K \longrightarrow \mathrm{R}f_*'^R g'^* K$$

to be the composite

$$\begin{aligned} g^*\mathrm{R}f_*^R K &\xrightarrow{\eta_{f'}^R} \mathrm{R}f_*'^R f'^* g^*\mathrm{R}f_*^R K \\ &\xrightarrow{\mathrm{R}f_*'^R(\theta_R)} \mathrm{R}f_*'^R g'^* f^*\mathrm{R}f_*^R K \\ &\xrightarrow{\mathrm{R}f_*'^R g'^*(\epsilon_f^R)} \mathrm{R}f_*'^R g'^* K. \end{aligned} \tag{8.2}$$

This is the right mate of the Cartesian inverse-image isomorphism (8.1)*.*

This definition makes clear why the compatibility of χ with both units and counits matters. We now prove the full exchange rectangle.

9 The coefficient-forgetting exchange rectangle

To keep the notation readable, write $\mathsf{U}_Z = \mathrm{Res}_{B/A,Z}$. We use the inverse-image identifications $\mathsf{U}g^* = g^*\mathsf{U}$ and $\mathsf{U}g'^* = g'^*\mathsf{U}$ without displaying λ.

Proposition 9.1 (Exchange rectangle)**.** *For every $F \in \mathrm{D}^+(X_{\text{ét}}, B)$, the following diagram commutes:*

$$\begin{array}{ccc} g^*\mathsf{U}_Y\mathrm{R}f_*^B F & \xrightarrow{\mathsf{U}_{Y'}(\beta_B(F))} & \mathsf{U}_{Y'}\mathrm{R}f_*'^B g'^* F \\ {\scriptstyle g^*(\chi_{f,F})}\downarrow & & \downarrow{\scriptstyle \chi_{f',g'^*F}} \\ g^*\mathrm{R}f_*^A\mathsf{U}_X F & \xrightarrow{\beta_A(\mathsf{U}_X F)} & \mathrm{R}f_*'^A g'^*\mathsf{U}_X F. \end{array} \tag{9.1}$$

Equivalently,

$$\chi_{f',g'^*F} \circ \mathsf{U}_{Y'}(\beta_B(F)) = \beta_A(\mathsf{U}_X F) \circ g^*(\chi_{f,F}). \tag{9.2}$$

Proof. We expand the upper route in (9.1) and transform it one piece at a time. From Definition 8.1,

$$\mathsf{U}(\beta_B(F)) = \mathsf{U}(\mathrm{R}f_*'^B g'^*(\epsilon_f^B)) \circ \mathsf{U}(\mathrm{R}f_*'^B(\theta_B)) \circ \mathsf{U}(\eta_{f'}^B). \tag{9.3}$$

Here and below the object subscripts on units are suppressed when they are forced by the source.

Step 1: move $\chi_{f'}$ through the two morphisms inside $\mathrm{R}f_*'$. Naturality (6.6) says that for every morphism $a : N \to N'$ on X', one has

$$\chi_{f',N'} \circ \mathsf{U}(\mathrm{R}f_*'^B a) = \mathrm{R}f_*'^A(\mathsf{U}a) \circ \chi_{f',N}. \tag{9.4}$$

Apply this first to $a = g'^*(\epsilon_f^B)$ and then to the Cartesian isomorphism $a = \theta_B$. The left side of (9.2) becomes

$$\mathrm{R}f_*'^A g'^*\mathsf{U}(\epsilon_f^B) \circ \mathrm{R}f_*'^A(\theta_A) \circ \chi_{f',,f'^*g^*\mathrm{R}f_*^B F} \circ \mathsf{U}(\eta_{f'}^B). \tag{9.5}$$

We used the fact that restriction of scalars carries θ_B to θ_A.

Step 2: replace the forgotten B-unit. Apply Proposition 7.1 to $u = f'$ and to the object $g^*\mathrm{R}f_*^B F$ on Y'. It gives

$$\chi_{f',,f'^*g^*\mathrm{R}f_*^B F} \circ \mathsf{U}(\eta_{f'}^B) = \eta_{f'}^A. \tag{9.6}$$

Substituting this into (9.5) yields

$$\mathrm{R}f'^{A}_{*}g'^{*}\mathsf{U}(\epsilon^{B}_{f})\circ\mathrm{R}f'^{A}_{*}(\theta_{A})\circ\eta^{A}_{f'}. \tag{9.7}$$

At this point the source of the A-unit is still $g^{*}\mathsf{U}_{Y}\mathrm{R}f^{B}_{*}F$.

Step 3: replace the forgotten B-counit. Proposition 7.2, applied to $u=f$, says

$$\mathsf{U}_{X}(\epsilon^{B}_{f})=\epsilon^{A}_{f}\circ f^{*}(\chi_{f,F}). \tag{9.8}$$

After applying g'^{*} and $\mathrm{R}f'_{*}$, expression (9.7) becomes

$$\mathrm{R}f'^{A}_{*}g'^{*}(\epsilon^{A}_{f})\circ\mathrm{R}f'^{A}_{*}g'^{*}f^{*}(\chi_{f,F})\circ\mathrm{R}f'^{A}_{*}(\theta_{A})\circ\eta^{A}_{f'}. \tag{9.9}$$

Step 4: move $g^{*}(\chi_{f,F})$ to the beginning. The Cartesian isomorphism is natural, so

$$g'^{*}f^{*}(\chi_{f,F})\circ\theta_{A}=\theta_{A}\circ f'^{*}g^{*}(\chi_{f,F}). \tag{9.10}$$

The unit $\eta^{A}_{f'}:\mathrm{id}\to\mathrm{R}f'_{*}f'^{*}$ is also natural, so

$$\mathrm{R}f'_{*}f'^{*}g^{*}(\chi_{f,F})\circ\eta^{A}_{f'}=\eta^{A}_{f'}\circ g^{*}(\chi_{f,F}). \tag{9.11}$$

Using (9.10) and (9.11), the composite (9.9) equals

$$\Big(\mathrm{R}f'^{A}_{*}g'^{*}(\epsilon^{A}_{f})\circ\mathrm{R}f'^{A}_{*}(\theta_{A})\circ\eta^{A}_{f'}\Big)\circ g^{*}(\chi_{f,F}). \tag{9.12}$$

The parenthesized composite is exactly $\beta_{A}(\mathsf{U}_{X}F)$ by (8.2). Therefore (9.12) is the right side of (9.2), which proves the proposition. □

10 Proof of the main theorem

We can now finish without repeating any geometric proper-base-change argument.

Proof of Theorem 1.1. Solving (9.2) for the forgotten B-coefficient map gives

$$\mathsf{U}_{Y'}(\beta_{B}(F))=\chi^{-1}_{f',g'^{*}F}\circ\beta_{A}(\mathsf{U}_{X}F)\circ g^{*}(\chi_{f,F}). \tag{10.1}$$

Every factor on the right is an isomorphism:

- $\chi_{f,F}$ and $\chi_{f',g'^{*}F}$ are isomorphisms by Proposition 6.1;
- an exact functor such as g^{*} carries $\chi_{f,F}$ to an isomorphism;
- $\beta_{A}(\mathsf{U}_{X}F)$ is an isomorphism by hypothesis.

Hence $\mathsf{U}_{Y'}(\beta_{B}(F))$ is an isomorphism in $\mathrm{D}^{+}(Y'_{\text{ét}},A)$. Proposition 3.4 says that $\mathsf{U}_{Y'}$ reflects isomorphisms. Therefore $\beta_{B}(F)$ is an isomorphism in $\mathrm{D}^{+}(Y'_{\text{ét}},B)$. □

Corollary 10.1 (Formal transfer of proper base change)**.** *Assume a proper-base-change theorem with A-coefficients asserts that $\beta_{A}(K)$ is an isomorphism for every relevant $K\in\mathrm{D}^{+}(X_{\text{ét}},A)$. Then for every $F\in\mathrm{D}^{+}(X_{\text{ét}},B)$, the B-coefficient map $\beta_{B}(F)$ is an isomorphism, as soon as the direct-image comparison* (6.1) *is available in the chosen coefficient formalism.*

11 Why the result is genuinely B-linear

One may worry that the argument proves only an isomorphism after forgetting the B-action. Conservativity already resolves this, but it is useful to see the B-linearity directly.

For a local section b of B and a B-complex F, multiplication defines an endomorphism

$$m_b : F \longrightarrow F.$$

Every functor and every unit, counit, and Cartesian isomorphism used in (8.2) is B-linear and natural in F. Therefore

$$\begin{array}{ccc} g^*\mathrm{R}f_*^B F & \xrightarrow{\beta_B(F)} & \mathrm{R}f_*'^B g'^* F \\ {\scriptstyle g^*\mathrm{R}f_*^B(m_b)}\downarrow & & \downarrow{\scriptstyle \mathrm{R}f_*'^B g'^*(m_b)} \\ g^*\mathrm{R}f_*^B F & \xrightarrow{\beta_B(F)} & \mathrm{R}f_*'^B g'^* F \end{array}$$

commutes. Thus $\beta_B(F)$ is a morphism in the B-linear derived category from the outset.

If its underlying A-linear morphism is invertible, its inverse is automatically B-linear. In an abelian category this follows from the calculation in Lemma 2.2; in the derived category it follows intrinsically from the cone argument in Proposition 3.4. Hence the conclusion is an isomorphism in $\mathrm{D}^+(Y'_{\text{ét}}, B)$, not merely in the category obtained by forgetting scalars.

12 What must be canonical, and what is insufficient

Warning 12.1 (An arbitrary A-linear isomorphism is not enough)**.** *Suppose one knows only that the two underlying A-coefficient objects*

$$g^*\mathrm{R}f_*^A(\mathrm{Res}\, F) \quad \textit{and} \quad \mathrm{R}f_*'^A g'^*(\mathrm{Res}\, F)$$

are isomorphic by some map γ. This alone says nothing about whether $\mathrm{Res}(\beta_B(F))$ *is an isomorphism. The map γ might be unrelated to the forgotten canonical exchange map.*

What is needed is either

1. *the assertion that the* canonical *map $\beta_A(\mathrm{Res}\, F)$ is an isomorphism, together with the exchange rectangle* (9.1)*; or*
2. *directly, the assertion that* $\mathrm{Res}(\beta_B(F))$ *is an isomorphism.*

The naturality of the mate construction is the bridge identifying the map in the A-coefficient theorem with the restriction of the B-coefficient map up to the comparison isomorphisms χ.

13 Scope, hypotheses, and common pitfalls

13.1 Where properness is used

Properness of f, torsion hypotheses, constructibility assumptions, or other geometric conditions belong to the theorem that proves $\beta_A(\mathrm{Res}\, F)$ is an isomorphism. Once that fact is known, the passage from A to B in Theorem 1.1 is formal and uses no new geometric argument.

13.2 Why the bounded-below condition appears

The composition theorem in Proposition 5.1 is stated for bounded-below derived categories and is proved using a bounded-below injective resolution together with the acyclic-resolution theorem. For unbounded derived categories, the same conclusion may still hold, but one must use an unbounded formalism, typically K-injective resolutions or a stable derived category, and verify that the coefficient comparison is compatible with that construction. One must not silently infer this from the bounded-below proof.

13.3 Constant coefficients versus a genuine change of ringed topos

This note uses constant coefficient sheaves $A_Z \to B_Z$ on every scheme and the ordinary inverse-image functor. If instead one studies a morphism of ringed topoi and defines pullback by

$$\mathcal{O}_{Z_1} \otimes_{u^{-1}\mathcal{O}_{Z_0}} u^{-1}(-),$$

then pullback need not be exact. Derived tensor products, Tor conditions, or finite Tor-dimension hypotheses may be required. As explained precisely in Remark 3.3, finite Tor amplitude controls whether a derived tensor functor preserves D^+; it is not needed for exactness of restriction of scalars.

13.4 Restriction is not extension

The implication proved here is

$$\begin{gathered}\text{the forgotten canonical map is an isomorphism}\\ \Longrightarrow\\ \text{the original } B\text{-linear map is an isomorphism.}\end{gathered}$$

It uses a conservative functor. Extension of scalars $B \otimes^{\mathrm{L}}_A -$ is generally not conservative and does not support the same argument without additional hypotheses.

13.5 A practical checklist

To reuse the proof in another six-functor setting, verify the following items explicitly:

1. The coefficient-forgetting functor is exact and conservative.
2. It commutes with the relevant inverse-image functors.
3. There is a natural isomorphism $\operatorname{Res}\mathrm{R}u^B_* \simeq \mathrm{R}u^A_* \operatorname{Res}$.
4. This isomorphism is the right mate of the inverse-image comparison, hence satisfies the unit and counit identities.
5. The exchange morphism is defined by the same unit–Cartesian-cell–counit formula in both coefficient categories.
6. The theorem with A-coefficients applies to the actual object $\operatorname{Res} F$ and to the canonical exchange map.

When these six points hold, no second proof of proper base change is needed.

A An abstract morphism-of-adjunctions lemma

The formal heart of the argument is independent of etale cohomology. We record it in a reusable form.

Proposition A.1 (Right mates preserve unit–counit constructions)**.** *Let*

$$L_B \dashv R_B, \qquad L_A \dashv R_A$$

be adjunctions between categories, and let $\mathsf{U}_0, \mathsf{U}_1$ *be functors for which there is a natural isomorphism*

$$\lambda : \mathsf{U}_1 L_B \xrightarrow{\sim} L_A \mathsf{U}_0.$$

Let $\chi : \mathsf{U}_0 R_B \to R_A \mathsf{U}_1$ *be the right mate of* λ*. Then:*

1. $\chi_{L_B M} \circ \mathsf{U}_0(\eta^B_M) = \eta^A_{\mathsf{U}_0 M}$ *after applying* λ*;*
2. $\mathsf{U}_1(\epsilon^B_K) = \epsilon^A_{\mathsf{U}_1 K} \circ L_A(\chi_K)$ *after applying* λ*;*
3. *every natural transformation constructed as a composite of a unit, a natural two-cell between left adjoints, and a counit is carried by* $(\mathsf{U}_0, \mathsf{U}_1)$ *to the corresponding* A*-transformation, conjugated by the relevant* χ *maps.*

Proof. The right mate is, by definition,

$$\mathsf{U}_0 R_B \xrightarrow{\eta^A} R_A L_A \mathsf{U}_0 R_B \xrightarrow{R_A \lambda^{-1}} R_A \mathsf{U}_1 L_B R_B \xrightarrow{R_A \mathsf{U}_1 \epsilon^B} R_A \mathsf{U}_1.$$

For the unit identity, precompose with $\mathsf{U}_0 \eta^B$ and use the triangle identity $R_B \epsilon^B \circ \eta^B R_B = \mathrm{id}_{R_B}$. After naturality of η^A and λ, the composite reduces to η^A.

For the counit identity, apply L_A to the displayed formula for χ and postcompose with ϵ^A. The triangle identity $\epsilon^A L_A \circ L_A \eta^A = \mathrm{id}_{L_A}$ cancels the inserted A-unit, leaving the transported B-counit.

For the third assertion, expand the given transformation into its three defining pieces. Use the unit identity on the first piece, naturality of λ on the middle two-cell, and the counit identity on the last piece. Naturality moves the remaining comparison maps to the outside of the composite. This is exactly conjugation by χ. □

B A compact proof template

For reference, the entire coefficient-transfer argument can be compressed to the following chain, with each equality justified in the main text:

$$\begin{gathered}
\mathrm{Res}_{Y'}(\beta_B(F)) = \chi^{-1}_{f',g'^*F} \circ \beta_A(\mathrm{Res}_X F) \circ g^*(\chi_{f,F}), \\
\beta_A(\mathrm{Res}_X F) \text{ is an isomorphism} \Longrightarrow \mathrm{Res}_{Y'}(\beta_B(F)) \text{ is an isomorphism}, \\
\mathrm{Res}_{Y'} \text{ conservative} \Longrightarrow \beta_B(F) \text{ is an isomorphism}.
\end{gathered}$$

The work lies in proving that the first line is an equality of the *canonical* maps. Sections 6, 7, and 9 provide precisely that verification.